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Faculty of Social Sciences

Dynamics of the International Trade Network in a Quasi-Static World-System

A Stochastic Blockmodel Comparison of 2015-2016 and 2019-2020

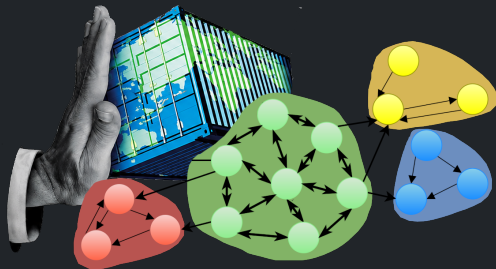
8th European Conference on Social Networks
August 14, 2026 || Norrköping 🇸🇪

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Agenda

1. Problem & Research questions
2. Data & Methods
3. Results
4. Conclusion

1.1 Renewing the ‘natural wedding’

- Orthodox international-trade economist have seldom used network methods (cf. [1]).
- Yet, there was a ‘natural wedding’ [2] between WSA and blockmodeling (BM) in the 1970s* [3]–[6].

Novi Ekonomist
Vol 17(1), Year XVII, Issue 33, January - June 2023.
ISSN 1840-2313 (Print) 2566-333X (Online)
DOI: 10.7251/NOEEN2333004T

Submitted: 24.04.2023.
Accepted: 19.06.2023.
Scientific article
UDK: 339.5/9(4-672EU):347.7(100)

ARE SANCTIONS FOR LOSERS? A NETWORK STUDY OF TRADE SANCTIONS

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Paper presented at the 11th International Scientific Symposium „EkonBiz: Impact of geopolitical changes on the national economy“, Brijuni, 22 – 23th June 2023.

American Journal of Sociology
Vol. 84, No. 5 (Mar., 1979),
pp. 1096-1126 (31 pages)



Structural Position in the World System and Economic Growth, 1955-1970: A Multiple-Network Analysis of Transnational Interactions

David Snyder and Edward L. Kick

- Deterministic BM can be sensitive to implementation and carries stringent assumptions (cf. [7]).
- It often relied on binarisation and/or pre-specified blockmodels.
- Renewed interest in network analyses of international trade (e.g.: [8], [9], [10]).

1.2 What is blockmodeling (BM)?

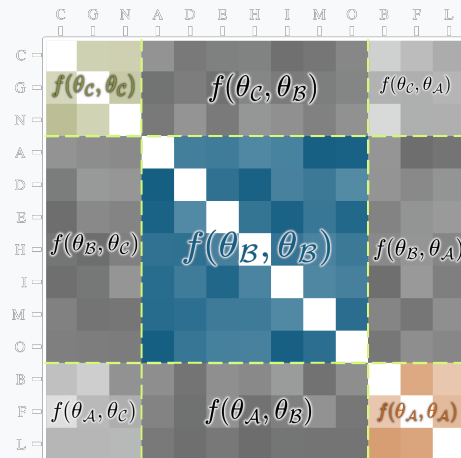


BM is an algebraic procedure that 'seeks clusters of equivalent units based on a selected definition of equivalence.' [11]

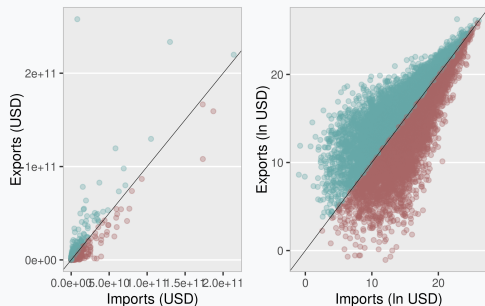
1.3 How does an SBM work?

- Given the adjacency matrix $[M]$, the ties m_{ij} are exclusively a function of the endpoints' latent-class membership (Z_i, Z_j) .
- Therefore, the model is assumed to be:

$$\mathbb{P}([M], \underline{Z}) = \mathbb{P}(\underline{Z}) \mathbb{P}([M] | \underline{Z})$$
- Estimation and optimisation typically through iterative methods (e.g., EM, split-merge, MCMC) to minimise an LL -based objective function (often the ICL)



1.4 The features of the ITN



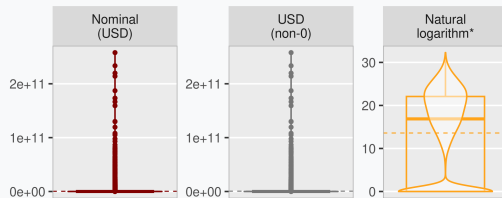
.. and why they are relevant to BM

- Trade is unbalanced → ITN should be **directed**
- Extensive vs intensive margins → ITN should be **weighted**.

⚠ **Binarisation** and **unit-pruning** can discard valuable information and create non-random missingness.

Breaking some of the SBM's assumptions

- **Nominal trade values** depend on country and dyadic covariates.
- Units have **unequal relational capacity** (URC).



1.5 Research questions

RQ1 Can the SBM recover a meaningful positional hierarchy?

Does an SBM fitted to the directed, weighted 2019-2020 ITN infer a positional hierarchy that can be meaningfully interpreted through the core/semi-periphery/periphery distinction?

RQ2 Is the blockmodel interpretable using WSA?

Do the inferred positions display inter- and intra-cluster flow patterns consistent with WSA expectations about asymmetric topological dependence?

Validation tasks

Sensitivity to SBM variant · Sensitivity to matrix normalisation · Temporal robustness

2.1 Dataset



UN Comtrade Database

- Direction: **Import** ($\Rightarrow$ CIF valuation).
- Year: **2019, 2020** (main period) and **2015, 2016** (temporal-robustness check)
- Product: **all traded goods**.

i Why use total trade?

Cons:

- Conceals countervailing commodity flows and product-composition effects.
- Cannot capture surplus-value transfer.
- Blockmodeling multiplex networks is possible (but computationally expensive).

Pros:

- Displays non-trivial higher-order properties [12].
- Findings often closely mirror those from commodity-specific networks [13].
- Better country coverage, higher-quality data.

2.2 Data pre-processing

Unit-filtering

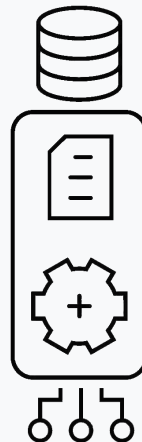
- *Goal:* Reduce noise, preserve broad country coverage
- *How:* LDCs and SIDSs with no reported ties.

Periodisation

- *Goal:* Smooth yearly shock and anomalies without cancelling them.
- *How:* Using two-year averages

Output

The ITN as a directed and weighted country-country matrix.



2.3 Normalisation

The Balassa Index (BI)

The SBMs are run on the Balassa-normalised matrix $[M]_{\text{BI}}$ [14]–[16], where ties are normalised by the ‘expected’ flow:

$$\text{BI}_{ij}^{(t)} = \frac{m_{ij}^{(t)} (\sum_{ij} m_{ij}^{(t)})}{(\sum_i m_{ij}^{(t)}) (\sum_j m_{ij}^{(t)})}, \text{ then } [M]_{\text{BI}} = \begin{cases} \ln \text{BI}_{ij} + \varepsilon, & \text{if } m_{ij} \neq 0 \\ 0, & \text{otherwise} \end{cases}$$

where ε is the minimum value that
ensures all logs are positive

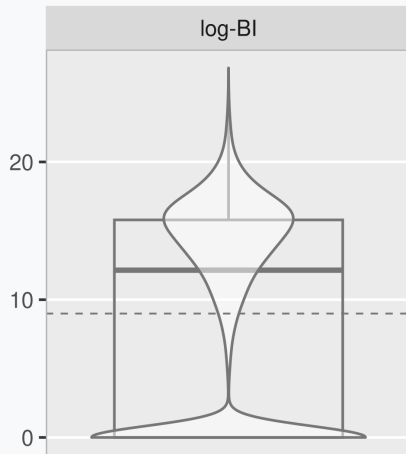
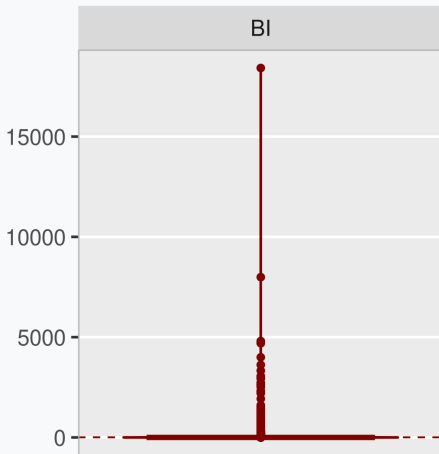


Usefulness of the BI for SBMs

- Reduces size effect and smooths URC on the intensive margins of trade.
- Aligns with WSA's focus on asymmetric dependence (unrelated to surplus transfer here).
- Still, does not solve the issue of URC on the extensive margins of trade.

2.3 Normalisation

Tie distribution using the BI



2.4 Methods

Main specification



- Hierarchical Bayesian stochastic blockmodel (HBSBM) as implemented in the Python/C++ library `graph-tool` [17]–[19].
- Estimates the blockmodel at multiple resolutions so smaller clusters can be identified in large networks and the selection of the partition's size is data-driven [17].
 - ⇒ Aligns with the view that positional analysis in WSA involves discretising a core-periphery continuum [20].
- Inference and MCMC-based optimisation aim at minimising the *minimum description length* (MDL), which is equivalent to maximising the LL [17].

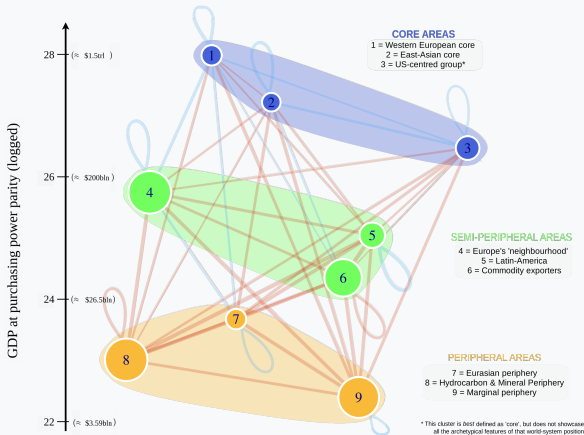
2.4 Methods

The other SBMs

	SBM variant			
	<i>SBM</i>	<i>DC SBM</i>	<i>DSBM</i>	<i>HBSBM</i>
<i>Formal characteristics</i>				
Equivalence	Stochastic	Stochastic	Stochastic	Stochastic
Objective function	ICL	MDL	ICL	MDL
Optimisation method	VEM	Greedy heuristic	VEM	Greedy heuristic
Partition-size selection	merge-split	MCMC Merge-split	Grid search	MCMC Merge-split
<i>Supported tie-value distributions</i>				
Gaussian	Yes	No	Yes	Yes
Poisson	Yes	Yes	Yes	Yes
Others	No	No	No	Yes
<i>Controls for unequal relational capacities (URC)</i>				
Degree-correction	No	Yes	No	Yes
Model for zeros	No	No	Yes	No

3.1 A nine-cluster partition

Overall structure



- *Core decoupling*: Western European core, East Asian core, US-centred group.
- The *semi-peripheries* are segmented by geography and GVCs.
- The *peripheral area* distinguishes weak embedding and sparseness.

3.1 A nine-cluster partition

Blockmodel analysis

	WE core	EA core	US group	EurNeigh	LatAm	CommExp	EurAsia	OilMineral	MargPer
WE core	0.44	-0.85	-1.12	0.03	-1.28	-0.82	-0.68	-2.22	-2.72
EA core	-1.23	0.36	-0.29	-1.74	-2.04	-1.06	-2.87	-1.76	-3.68
US group	-0.48	-0.17	-0.65	-1.16	-1.48	-1.73	-2.42	-3.2	-3.19
EurNeigh	0.06	-1.18	-1.86	0.01	-2.51	-2.27	-0.47	-5.07	-4.65
LatAm	-1.11	-0.72	-1.92	-1.93	1	-4.26	-4.04	-7.42	-4.47
CommExp	-0.93	-0.42	-0.89	-1.73	-4.06	-2.32	-4.33	-4.32	-7.07
EurAsia	-0.46	-1.25	-2.01	-0.29	-2.87	-2.83	-0.45	-6.23	-5.79
OilMineral	-0.75	-0.52	-0.8	-1.74	-4	-2.48	-4.09	-4.58	-19
MargPer	-1.85	-1.31	-2.29	-3.53	-3.19	-7.84	-8.89	-19	-8.19

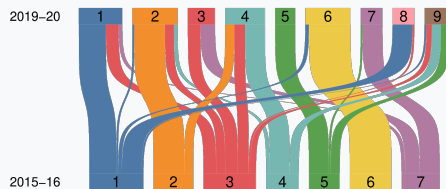
- **Core** clusters show less non-preferential (positive or less-negative) trade.
- The **semi-peripheries'** topological dependence¹ appears as sparsity+over-orientation.
- The *peripheral areas* see more negative log-BIs and lower nominal flows.

¹Block-level asymmetries are not direct proof of country-level dependency.

3.2 Robustness & sensitivity

- *Why is the HBSBM 'better'?* The standard SBM and DSBM recover **some common structure**, but reproduced an **activity gradient** with a very large core-like group. The HBSBM retains relational differences (e.g., separating NIG/KAZ).
- *Why is BI 'better'?* Nominal flows with DC produce **larger groups** driven more by **absolute trade value**, whereas BI emphasises relative orientation (see Vietnam).
- *What about a gravity model?* Residuals accentuates **exceptional dyads/outliers**, but the structure is less interpretable.

A quasi-static hierarchy



Conclusions

SBMs *can* for WSA

- The HBSBM partition can be interpreted through WSA.
- Using the BI allows to analyse the ITN in terms of over-/under-orientation and concentration/disconnection.
- Interpretability is conditional on defensible choices about network construction, data-preprocessing, and model specification.

Future work

Future work should investigate dependency and surplus-value transfer using:

- multi-relational and properly dynamic SBMs;
- data on value added (not gross CIF/FOB valuations);
- referents other than nation-states.

Thank you for you attention

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